Geography Ch-5 (Advanced) — India's Drainage System & River Interlinking

Advanced / India-applied | GS-I & GS-III | Source: D.R. Khullar + Majid Husain (India geography)
Himalayan vs Peninsular drainage, antecedent rivers, river capture, evolution
Extra CA: Ken-Betwa river interlinking (distinct from Must-Do's delta CA)

Facts from official sources | Inferences clearly marked | No fabricated data

■ HIGH UPSC RELEVANCE

HIGH

ADVANCED — Himalayan vs Peninsular Drainage & Antecedent Rivers (Khullar / Majid)

GS-I · India Geography → Drainage (Source: D.R. Khullar, Majid Husain)

■ NEWS TRIGGER

India's drainage splits into the **Himalayan** (extra-Peninsular) and **Peninsular** systems — a distinction that governs perennial vs seasonal flow, hydropower, floods and inter-basin transfer debates.

■ INTRO & ORIGIN

The **Himalayan drainage** (Indus, Ganga, Brahmaputra) is largely **perennial**, fed by glaciers, springs and rain. The **Peninsular drainage** is older, **seasonal**, and flows in near-fixed valleys.

*Himalayan rivers are largely **antecedent** — older than the mountains, they kept cutting down as the Himalaya rose, carving deep gorges. Khullar traces their evolution via the hypothetical **Indo-Brahm (Siwalik) river** and its successive **river captures** (headward erosion) by the Indus, Punjab rivers and Dihang.*

■ KEY DATA

Feature	Himalayan drainage	Peninsular drainage
Flow	Perennial (glacier + rain fed)	Seasonal (rain fed)
Age / valley	Young, deep gorges, meandering	Old, shallow, graded valleys
Type	Antecedent + consequent	Mostly consequent, super-imposed
Examples	Indus, Ganga, Brahmaputra	Godavari, Krishna, Mahanadi, Kaveri

■ STATIC THEORY

• The **Himalayan drainage** comprises India's international rivers — Indus, Ganga, Brahmaputra — mostly perennial.

• **Antecedent drainage:** rivers predating the Himalaya that maintained their course by down-cutting as the range uplifted (deep transverse gorges).

• **River capture (piracy):** a more vigorous river, through **headward erosion**, beheads a neighbouring river — a key process in Himalayan drainage evolution.

• **Indo-Brahm hypothesis (Khullar):** a mighty former river along the Himalayan foredeep was dismembered by successive captures into today's Indus, Ganga and Brahmaputra systems.

• **Peninsular rivers** flow in comparatively fixed, shallow valleys; most drain east to the Bay of Bengal (except Narmada & Tapi, which flow west via rift valleys to the Arabian Sea).

■ MUST-KNOW FACTS

1. Himalayan = perennial, antecedent; Peninsular = seasonal, older.

2. Antecedent river = older than the mountain it cuts across (deep gorge).

3. River capture driven by headward erosion (Khullar: Indo-Brahm captures).

4. Peninsular rivers mostly flow EAST; Narmada & Tapi flow WEST (rift valleys).

■■ UPSC TRAPS

✗ All Peninsular rivers flow east to the Bay of Bengal.

✓ Narmada and Tapi are exceptions — they flow WEST through rift valleys to the Arabian Sea.

✗ Antecedent and consequent rivers are the same.

✓ Antecedent predates the uplift (cuts across it); consequent follows the initial slope after uplift.

5. Indus, Ganga, Brahmaputra = India's three international/Himalayan systems.

✗ Himalayan rivers are seasonal like Peninsular ones.

✓ Himalayan rivers are largely PERENNIAL (glacier + rain fed); Peninsular are seasonal (rain fed).

■ MAINS ANGLE

Contrast the origin and character of Himalayan and Peninsular drainage, and explain how antecedent drainage and river capture shaped today's river systems.

■ STUDY LINK

Geography → India → Drainage System (Khullar: Evolution of Himalayan Rivers; Majid: River Systems)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA ANCHOR — Ken-Betwa River Interlinking Project

GS-I / GS-III · India drainage applied → Inter-basin transfer

■ NEWS TRIGGER

Ken-Betwa is the FIRST project under India's National Perspective Plan for river interlinking — transferring surplus Ken (Bundelkhand, MP) water to the Betwa basin (UP) for irrigation, drinking water and power.

■ INTRO & ORIGIN

River interlinking applies drainage-basin logic to move 'surplus' water to deficit basins — but crosses ecological and social thresholds, making it a live GS-I geography + GS-III environment debate.

■ STATIC THEORY

• Objective: irrigate ~10+ lakh hectares, supply drinking water and hydropower in drought-prone **Bundelkhand**.

• Concerns: submergence of part of **Panna Tiger Reserve**, forest/biodiversity loss, displacement, high cost (~Rs 44,605 crore).

• Institutions: **Central Water Commission**, Ken-Betwa Link Project Authority; other links for comparison — Par-Tapi-Narmada, Damanganga-Pinjal.

■ MUST-KNOW FACTS

1. Ken-Betwa = 1st river-interlinking project (National Perspective Plan).

2. Ken (MP) surplus → Betwa (UP); Bundelkhand drought mitigation.

3. Key concern: Panna Tiger Reserve submergence + biodiversity loss.

■■ UPSC TRAPS

✗ Ken-Betwa transfers Betwa water into the Ken.

✓ It transfers surplus KEN water TO the Betwa basin.

■ MAINS ANGLE

Use drainage-basin geography to weigh inter-basin transfer benefits vs ecological costs (Panna).

■ STUDY LINK

Geography → India Drainage → Inter-basin Transfer (applied CA)

End of Register Notes — UPSC Agent / Copilot CLI